

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mining Engineering

2. Subject Name: Mining Machinery and Automation
L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: Solid Mechanics

4. Syllabus and reference books:

Syllabus:

Overview of Mining Machinery, Basics: Linkage, Belt, Rope, Chain, Gear, Flywheel, Governors, Brakes and Dynamometers, Coupling, Clutch, Cam and Follower, Power Transmission: IC Engines, Automobile Engineering.

Fundamentals of robotic systems: Algorithmic approach to robotics, Kinematics, dynamics, motion planning and control, Multi-link robot arms and mobile robot systems.

Machineries for Mining (Conventional, Hydraulic and Pneumatic and Unmanned): Shovel, Dragline, Bucket Wheel Excavator, Front end Loader, Scraper, Dumpers, Dozer, Grader, Ripper, Rollers, Continuous Miner, Side Discharge Loader, Load Haul Dump Machine, Shearers.

Auxiliary Equipment and Operations: Drilling Machine, Pump, Flow Control Valve, Mine Drainage, Transportation Machineries: Belt Conveyors, Slurry Transport, Underground Transport Systems; Winning machines for surface and underground: Underground man riding systems, Wire ropes and winding systems.

Maintenance management for mining machinery, Machineries for Tunneling: Tunnel Boring Machines, Roadheaders.

Reference Books:

- 1) Khurmi, R.S. and Gupta J.K. (1976). Theory of Machines. S.Chand Publisher.
- 2) Ganesan, V. (2018). Internal Combustion Engines, 4th Edition, McGraw Hill Education Pvt. Ltd.
- 3) Pathak, K. (2009). Mining Machinery Maintenance and Capacity Utilization. Cygnus Publisher.
- 4) De, A. (2014). Latest Development of Heavy Earth Moving Machinery. Galgotias Publisher.
- 5) Walker, S.C. (1988). Mine Winding and Transport. Elsevier.
- 6) Ramlu, M.A. (2015). Mine Hoisting, White Falcon Publishing

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Overview of Mining Machinery, Basics: Linkage, Belt, Rope, Chain, Gear, Flywheel, Governors, Brakes and Dynamometers, Coupling, Clutch, Cam and Follower, Power Transmission: IC Engines, Automobile Engineering.	12
2.	Fundamentals of robotic systems: Algorithmic approach to robotics, Kinematics, dynamics, motion planning and control, Multi-link robot arms and mobile robot systems.	4
3.	Machineries for Mining (Conventional, Hydraulic and Pneumatic and Unmanned): Shovel, Dragline, Bucket Wheel Excavator, Front end Loader, Scraper, Dumpers, Dozer, Grader, Ripper, Rollers, Continuous Miner, Side Discharge Loader, Load Haul Dump Machine, Shearers.	8
	Auxiliary Equipment and Operations: Drilling Machine, Pump, Flow Control Valve, Mine Drainage, Transportation Machineries: Belt Conveyors, Slurry Transport, Underground Transport Systems; Winning machines for surface and underground: Underground man riding systems, Wire ropes and winding systems.	6
5.	Maintenance management for mining machinery, Machineries for Tunneling: Tunnel Boring Machines, Roadheaders.	6
Total number of hours		36